



EXTENDED EUCLIDS WITH BEZOUTS COEFFICIENT

Problem Submissions Leaderboard

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Euclid's Identity states that for any two positive integers A and B, there exist integers x and y such that: $Ax + By = D$ where D is the Greatest Common Divisor (GCD) of A and B. Given two positive integers A and B, find: $x \ y \ D = \text{gcd}(A, B)$ such that:

$Ax + By = D$

If multiple pairs (x, y) satisfy the equation, choose the pair with the minimum value of: $|x| + |y|$

If there is still more than one pair, choose the pair for which:

$x \leq y$

Input Format

A B

Constraints

$1 \leq A, B \leq 10^9$

Output Format

$x \ y \ D$ D is $\text{gcd}(A, B)$ $Ax + By = D$ (x, y) satisfies the minimum-selection rules.

Sample Input 0

4 6

Sample Output 0

-1 2

Change Theme

Language

Python 3



```
1 def extended_gcd(a, b):
2     if b == 0:
3         return a, 1, 0
4     g, x1, y1 = extended_gcd(b, a % b)
5     x = y1
6     y = x1 - (a // b) * y1
7
8     return g, x, y
9
10 def solve(A, B):
11     D, x0, y0 = extended_gcd(A, B)
12
13     p = B // D
14     q = A // D
```

```
15
16     candidates = set()
17
18     k1 = -x0 / p
19
20     k2 = y0 / q
21
22     for k_real in (k1, k2):
23         k = int(k_real)
24
25         for candidate_k in range(k - 2, k + 3):
26             candidates.add(candidate_k)
27
28     best = None
29
30     for k in candidates:
```

Line: 52 Col: 1

 Upload Code as File

☐ Test against custom input

Run Code

Submit Code

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✔ Sample Test case 0

Input (stdin)

Download

1 4 6

Your Output (stdout)

1 -1 1 2

Expected Output

Download

1 -1 1 2